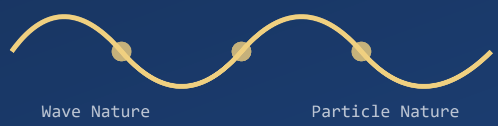
B.TECH – 1ST YEAR | ENGINEERING PHYSICS

Engineering Physics

UNIT IV

Quantum Mechanics &
Free Electron Theory



Comprehensive Study Module

Includes: Derivations · Diagrams · Q&A · Quick Revision

Based on: Avadhanulu & Kshirsagar | Bhattacharya & Tandon

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1 Introduction to Quantum Mechanics

Definition: Quantum Mechanics is the branch of physics that deals with the behaviour of matter and energy at the atomic and subatomic level, where classical Newtonian mechanics fails to describe the phenomena accurately.

1.1 Historical Background

Classical physics, developed over centuries by Newton, Maxwell, and others, explained macroscopic phenomena with great precision. However, by the late 19th century, several experimental observations could not be explained by classical theories:

- Blackbody radiation spectrum – explained by **Max Planck (1900)** using energy quantization: $E = h\nu$
- Photoelectric effect – explained by **Einstein (1905)** using photons: $E = h\nu - \phi$
- Atomic line spectra – explained by **Bohr (1913)** using quantized orbits
- Compton scattering – confirmed photon momentum: $p = h/\lambda$
- Davisson–Germer experiment – proved electron diffraction (wave nature of particles)

1.2 Planck's Quantum Hypothesis

Planck proposed that energy is not continuous but is emitted or absorbed in discrete packets called **quanta**. The energy of one quantum is:

PLANCK'S ENERGY QUANTUM

$$E = h\nu$$

$h = 6.626 \times 10^{-34}$ J·s (Planck's constant), ν = frequency of radiation

$$E = \hbar\omega$$

$\hbar = h/2\pi$ (reduced Planck's constant), ω = angular frequency

1.3 Photoelectric Effect – Einstein's Contribution

Einstein extended Planck's idea to explain that light consists of particles called **photons**, each carrying energy $E = h\nu$. When a photon strikes a metal surface, it ejects an electron if its energy exceeds the work function φ:

EINSTEIN'S PHOTOELECTRIC EQUATION

$$h\nu = \phi + \tfrac{1}{2}mv^2_{max}$$

$$h\nu = h\nu_0 + eV_s$$

φ = work function, ν_0 = threshold frequency, V_s = stopping potential

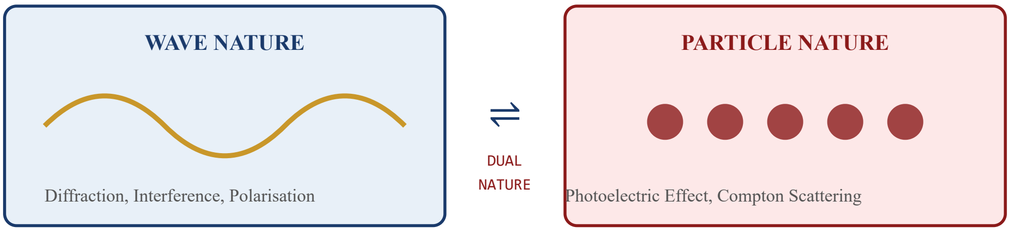

Fig. 1.1 — Dual Nature of Radiation: Wave & Particle Behaviour

1.4 Need for Quantum Mechanics

- Classical mechanics fails at atomic/subatomic scales
- Quantization of energy and angular momentum observed experimentally
- Wave–particle duality demands a new mathematical framework
- Schrödinger's wave equation provides the governing law of quantum systems

2 Dual Nature of Matter (de Broglie Hypothesis)

de Broglie Hypothesis (1924): Every moving particle, whether microscopic or macroscopic, has an associated matter wave. The wavelength of this wave (called the de Broglie wavelength) is given by $\lambda = h/p$, where p is the momentum of the particle.

2.1 de Broglie's Reasoning

de Broglie argued by analogy: if radiation (classically a wave) has particle-like behaviour (photons), then particles (classically matter) should exhibit wave-like behaviour. He used Einstein's mass–energy equivalence and Planck's quantum relation:

1. For a photon: $E = h\nu = hc/\lambda$
2. From Einstein's relation: $E = mc^2$, so $mc = h\nu/c = h/\lambda$
3. Photon momentum: $p = mc = h/\lambda$, hence $\lambda = h/p$
4. By analogy, for a matter particle of mass *m* moving with velocity *v*: $p = mv$
5. Therefore, de Broglie wavelength: $\lambda = h/mv$

DE BROGLIE WAVELENGTH FORMULAS

$$\lambda = h/p = h/mv$$

General form; p = mv = momentum

$$\lambda = h/\sqrt{(2mE_k)}$$

E_k = kinetic energy = ½mv²

$$\lambda = h/\sqrt{(2meV)}$$

For electron accelerated through potential V

$$\lambda = 12.27/\sqrt{V}\ \text{Å}$$

Simplified for electrons (V in volts)

2.2 Experimental Verification — Davisson–Germer Experiment (1927)

- Electrons were directed onto a nickel crystal surface
- Diffraction pattern was observed, similar to X-ray diffraction
- The experimentally measured wavelength matched de Broglie's formula
- Confirmed: electrons exhibit wave nature
- Also independently confirmed by G.P. Thomson (electron diffraction through thin foil)

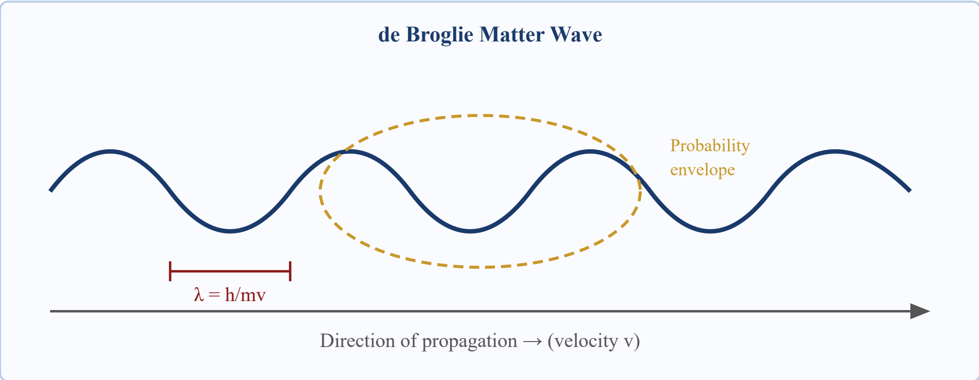

Fig. 2.1 — de Broglie Matter Wave: Wave packet associated with a moving particle

2.3 Properties of de Broglie Waves

- Not electromagnetic waves; they are probability waves (ψ-waves)
- Wavelength decreases as momentum increases ($\lambda \propto 1/p$)
- For macroscopic objects, λ is negligibly small → quantum effects imperceptible
- Phase velocity of de Broglie wave: $v_p = E/p = c^2/v$ (always > c for v < c)
- Group velocity equals particle velocity: $v_g = v$

2.4 Merits of de Broglie Hypothesis

✓ Merits

- Explains Bohr's quantization condition: $2\pi r = n\lambda = nh/mv \rightarrow mvr = n\hbar$
- Provides basis for Schrödinger's wave equation
- Explains electron diffraction and interference
- Led to development of electron microscope (sub-Å resolution)

2.5 Applications

- **Electron Microscope:** Uses de Broglie wavelength of electrons (~0.01 Å) for ultra-high-resolution imaging
- **Neutron Diffraction:** Used for crystal structure analysis
- **Quantum Tunnelling:** Basis for tunnel diodes, STM (scanning tunnelling microscope)

3 Heisenberg's Uncertainty Principle

Statement: It is fundamentally impossible to determine simultaneously and precisely both the position and momentum of a quantum particle. The product of uncertainties in position (Δx) and momentum (Δp) is always greater than or equal to $\hbar/2$.

HEISENBERG'S UNCERTAINTY RELATIONS

$$\Delta x \cdot \Delta p_x \geq \hbar/2$$

Position–momentum uncertainty (most common form)

$$\Delta x \cdot \Delta p_x \geq h/(4\pi)$$

Since $\hbar = h/2\pi$, alternate form

$$\Delta E \cdot \Delta t \geq \hbar/2$$

Energy–time uncertainty relation

$$\Delta L \cdot \Delta \theta \geq \hbar/2$$

Angular momentum–angle uncertainty

3.1 Physical Significance

- This is NOT due to experimental limitations but a fundamental property of nature
- The more precisely position is known, the less precisely momentum is known, and vice versa
- Arises from the wave nature of particles — a perfectly localized wave packet requires infinite spread in momentum space
- Explains why electrons cannot exist inside the nucleus (confinement to nucleus would require impossibly large kinetic energy)

3.2 Proof by Single-Slit Diffraction

1. Consider an electron passing through a slit of width Δy. The position uncertainty: `Δy = d (slit width)`

2. Due to diffraction, the electron's transverse momentum Δp_y is uncertain. From first minimum of diffraction: $\sin\theta = \lambda/d$

3. Transverse momentum: $\Delta p_y = p \sin\theta = (h/\lambda)(\lambda/d) = h/d$

4. Therefore: $\Delta y \cdot \Delta p_y = d \cdot (h/d) = h \approx \hbar$ (order of magnitude)

5. This confirms the uncertainty principle from wave diffraction

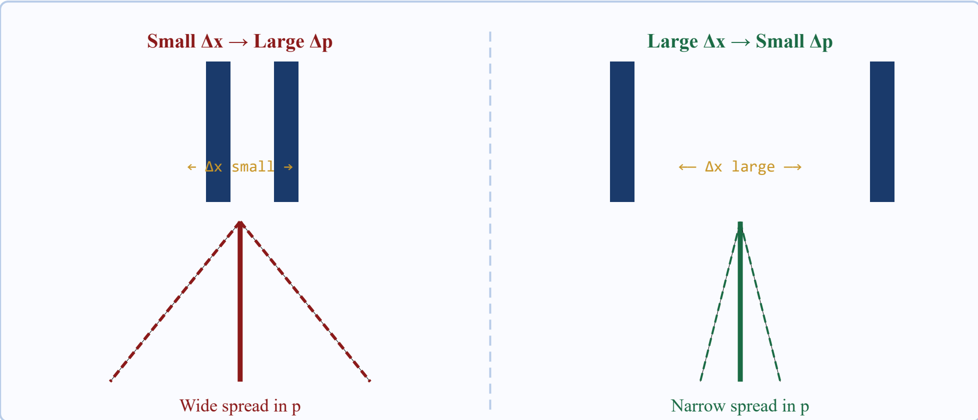

Fig. 3.1 — Uncertainty Principle: Single-slit diffraction analogy

3.3 Applications of Uncertainty Principle

- **Non-existence of electrons in nucleus:** Confinement to $\Delta x \approx 10^{-14}$ m requires $\Delta p \approx 10^{-20}$ kg·m/s → KE ≈ 20 MeV, far exceeding observed β-particle energies
- **Zero-point energy:** Quantum particle in a box cannot be at rest; minimum energy = $h^2/(8mL^2)$
- **Width of spectral lines:** $\Delta E \cdot \Delta t \geq \hbar$ → finite lifetime Δt → finite energy spread ΔE → line broadening
- **Stability of atoms:** Electron cannot spiral into nucleus as that would violate uncertainty principle

⚠ **Note:** The Uncertainty Principle is NOT a statement about measurement error. It reflects the intrinsic, irreducible quantum nature of reality itself. Even with perfect instruments, $\Delta x \cdot \Delta p \geq \hbar/2$ always holds.

4 Wave Function and Its Properties

Wave Function (ψ): A mathematical function $\psi(x, y, z, t)$ that completely describes the quantum state of a particle. It is a complex-valued function whose squared modulus $|\psi|^2$ gives the probability density of finding the particle at a given location and time.

4.1 Physical Interpretation — Born's Interpretation

Max Born (1926) proposed the statistical interpretation of the wave function:

PROBABILITY DENSITY

$$P(x, t) = |\psi(x, t)|^2 = \psi \cdot \psi^*$$

Probability density (real, non-negative)

$$P(x, t)\, dx = |\psi|^2\, dx$$

Probability of finding particle in [x, x+dx]

$$\int_{-\infty}^{\infty} |\psi|^2\, dx = 1$$

Normalization condition (particle must be somewhere)

4.2 Properties of a Valid Wave Function

- **Single-valued:** ψ must have only one value at each point in space (unique probability at each location)
- **Finite:** ψ must be finite everywhere (probability cannot be infinite)
- **Continuous:** ψ and its first derivatives must be continuous (no abrupt jumps)
- **Normalizable:** $\int|\psi|^2\, dV$ must be finite, so the function vanishes at infinity
- **Square-integrable:** The total probability over all space must equal 1
- **Smoothly varying:** $\partial\psi/\partial x$, $\partial\psi/\partial y$, $\partial\psi/\partial z$ must all be continuous

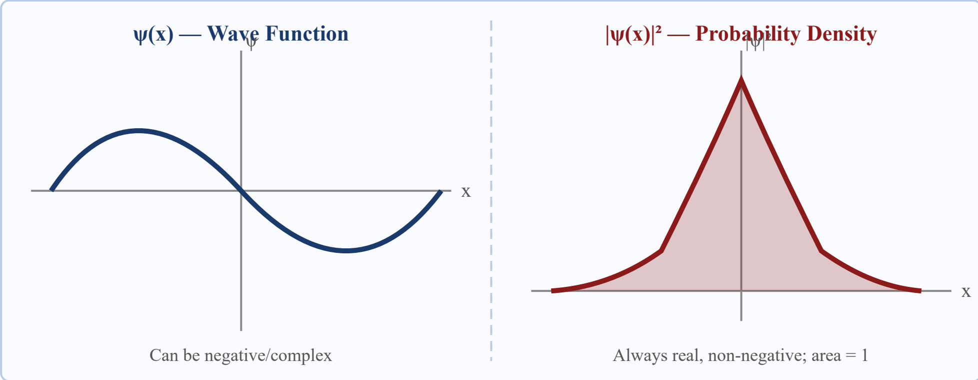

Fig. 4.1 — Wave Function ψ(x) and Probability Density |ψ(x)|²

4.3 Expectation Values

The average (expected) value of any observable quantity A is computed as:

EXPECTATION VALUE

$$\langle x \rangle = \int_{-\infty}^{\infty} \psi^* \, x \, \psi \, dx$$

Average position

$$\langle p \rangle = \int_{-\infty}^{\infty} \psi^* \left(-i\hbar \, \partial/\partial x\right) \psi \, dx$$

Average momentum (using momentum operator)

4.4 Significance of Wave Function

- Encodes all information about the quantum state of the system
- $|\psi|^2$ gives measurable probability densities
- Obeys the superposition principle: $\psi = c_1\psi_1 + c_2\psi_2$ is also a valid wave function
- Evolves deterministically via Schrödinger's equation between measurements
- Collapses probabilistically upon measurement (Copenhagen interpretation)

5 Schrödinger Wave Equations

Schrödinger's Wave Equation is the fundamental equation of quantum mechanics, analogous to Newton's second law in classical mechanics. It governs how the wave function of a quantum system evolves in space and time.

5.1 Time-Dependent Schrödinger Equation (TDSE)

Derivation

1. Start with the de Broglie wave for a free particle moving in x-direction: $\psi(x, t) = A\, e^{i(kx - \omega t)}$ where $k = p/\hbar,\ \omega = E/\hbar$
2. Differentiate with respect to time: $\partial\psi/\partial t = -i\omega\, \psi = -i(E/\hbar)\, \psi$
3. Therefore: $i\hbar\, \partial\psi/\partial t = E\, \psi$ → This identifies the energy operator: $\hat{E} = i\hbar\, \partial/\partial t$
4. Differentiate twice w.r.t. x: $\partial^2\psi/\partial x^2 = -k^2\, \psi = -(p^2/\hbar^2)\, \psi$
5. So: $p^2\, \psi = -\hbar^2\, \partial^2\psi/\partial x^2$ → Momentum operator: $\hat{p} = -i\hbar\, \partial/\partial x$
6. Total energy: $E = p^2/2m + V(x, t)$
7. Apply energy on ψ: $E\, \psi = (p^2/2m + V)\, \psi = -(\hbar^2/2m)\, \partial^2\psi/\partial x^2 + V\, \psi$
8. Combining steps 3 and 7, the TDSE in 1D:

TIME-DEPENDENT SCHRÖDINGER EQUATION (1D)

$$i\hbar\, \partial\psi/\partial t = -(\hbar^2/2m)\, \partial^2\psi/\partial x^2 + V(x,t)\, \psi$$

$\hbar = h/2\pi$, m = mass, V = potential energy

$$i\hbar\, \partial\psi/\partial t = \hat{H}\, \psi$$

$\hat{H}$ = Hamiltonian operator = $-(\hbar^2/2m)\, \nabla^2 + V$

$$\text{In 3D: } i\hbar\, \partial\psi/\partial t = -(\hbar^2/2m)\nabla^2\psi + V\, \psi$$

$\nabla^2 = \partial^2/\partial x^2 + \partial^2/\partial y^2 + \partial^2/\partial z^2$ (Laplacian)

5.2 Time-Independent Schrödinger Equation (TISE)

When the potential V depends only on position (not time), the wave function separates:

1. Assume: $\psi(x, t) = \phi(x) \cdot T(t)$ (separation of variables)
2. Substitute in TDSE: $i\hbar\, \phi\, dT/dt = T\, [-\hbar^2/2m \cdot d^2\phi/dx^2 + V\, \phi]$
3. Divide throughout by $\psi = \varphi T$: $i\hbar\, (1/T)\, dT/dt = (1/\phi)[-\hbar^2/2m \cdot d^2\phi/dx^2 + V\, \phi] = E$ (separation constant)
4. Time equation: $T(t) = e^{-iEt/\hbar}$ (gives time-oscillating factor)
5. Spatial equation (TISE): $-(\hbar^2/2m)\, d^2\phi/dx^2 + V\, \phi = E\, \phi$

TIME-INDEPENDENT SCHRÖDINGER EQUATION (TISE)

$$-(\hbar^2/2m)\, d^2\psi/dx^2 + V(x)\, \psi = E\, \psi$$

One-dimensional TISE

$$d^2\psi/dx^2 + (2m/\hbar^2)(E - V)\, \psi = 0$$

Rearranged form; $k^2 = 2m(E-V)/\hbar^2$

$$\nabla^2\psi + (2m/\hbar^2)(E - V)\, \psi = 0$$

Three-dimensional TISE

5.3 Physical Significance

- TDSE: Governs the complete evolution of quantum state with time
- TISE: Applies to stationary states (constant potential); gives allowed energy levels
- Solutions of TISE are called **eigenfunctions**; corresponding values E are **eigenvalues**
- Equation is linear ⟶ superposition principle holds
- Replaces Newton's F = ma for quantum particles

6 Particle in a One-Dimensional Infinite Potential Well

Particle in a Box: A model quantum system consisting of a particle confined within a region of length L by infinitely high potential walls. It is the simplest quantum mechanical problem and illustrates energy quantization.

6.1 Potential Description

The potential energy is defined as:

POTENTIAL FUNCTION

$$V(x) = 0, \text{ for } 0 \le x \le L$$

$$V(x) = \infty, \text{ for } x < 0 \text{ or } x > L$$

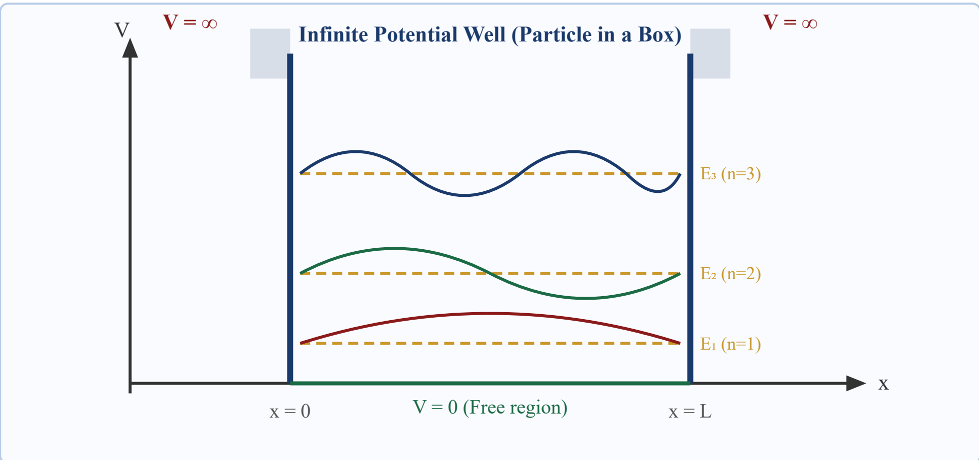

Fig. 6.1 — Infinite Potential Well showing wave functions ψ_n and energy levels E_n

6.2 Solution of Schrödinger Equation

1. Inside well (V = 0), TISE becomes: $d^2\psi/dx^2 + k^2\psi = 0$ where $k^2 = 2mE/\hbar^2$
2. General solution: $\psi(x) = A\sin(kx) + B\cos(kx)$
3. Boundary condition at x = 0: $\psi(0) = 0 \rightarrow B = 0 \rightarrow \psi = A\sin(kx)$

4 Boundary condition at x = L: $\psi(L) = 0 \rightarrow A \sin(kL) = 0 \rightarrow kL = n\pi$

5 So: $k = n\pi/L$, and since $k^2 = 2mE/\hbar^2$: $E = n^2\pi^2\hbar^2/2mL^2 = n^2h^2/8mL^2$

6 Normalization: $\int_0^L |\psi|^2 \, dx = 1 \rightarrow A = \sqrt{(2/L)}$

RESULTS FOR PARTICLE IN A BOX

$$\psi_n(x) = \sqrt{(2/L)} \cdot \sin(n\pi x/L)$$

Normalized wave function; n = 1, 2, 3, ...

$$E_n = n^2h^2/(8mL^2) = n^2\pi^2\hbar^2/(2mL^2)$$

Allowed energy levels (quantized!)

$$E_n = n^2 E_1$$

$E_1 = h^2/8mL^2$ = ground state (zero-point) energy

6.3 Energy Level Diagram

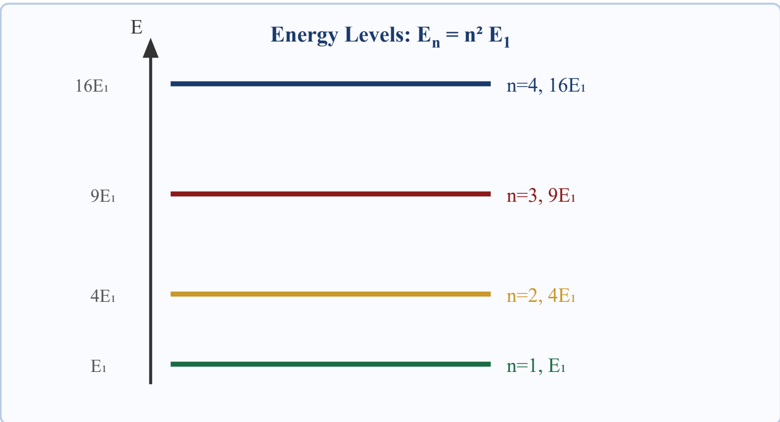

Fig. 6.2 — Quantized Energy Levels in Infinite Potential Well

6.4 Key Observations

- **Energy is quantized:** Only discrete values $E_n = n^2 E_1$ are allowed
- **Zero-point energy:** Even for n = 1, $E_1 \neq 0$ (particle cannot be at rest)
- **n nodes:** ψ_n has (n−1) nodes inside the well
- Energy gaps increase with n: E levels spread further apart as n increases

7 Introduction to Free Electron Theory

Free Electron Theory: A model that treats conduction electrons in a metal as free particles (not bound to individual atoms), which can move freely throughout the metallic lattice. This model explains electrical and thermal conductivity.

7.1 Background

Metals are excellent conductors of electricity and heat. In metallic bonding, valence electrons detach from their parent atoms and form a "sea" or "gas" of free electrons that can move throughout the lattice. Two major models have been developed:

Aspect	Classical Free Electron Theory	Quantum Free Electron Theory
Proposed by	Drude (1900), Lorentz (1909)	Sommerfeld (1928)
Statistics used	Maxwell–Boltzmann	Fermi–Dirac
Energy distribution	Continuous (classical)	Quantized (discrete levels)
Electron–electron interaction	Ignored	Ignored (but Pauli exclusion applied)
Success	Ohm's Law, Wiedemann–Franz	Specific heat, photoelectric, thermionic

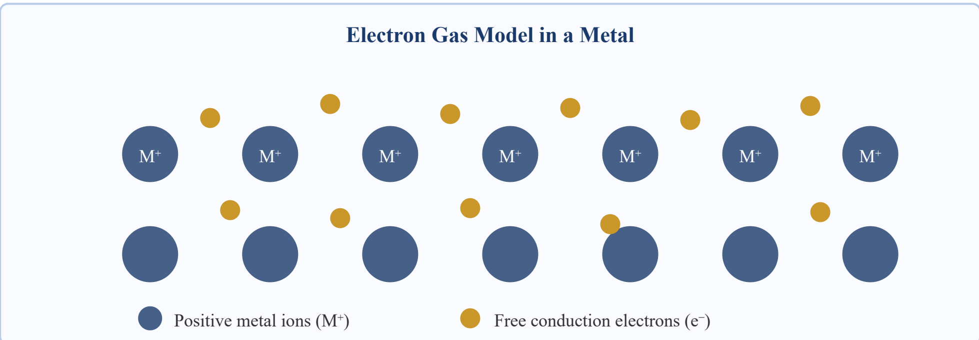

Fig. 7.1 — Free Electron (Electron Gas) Model of a Metal

8 Classical Free Electron Theory (Drude–Lorentz Model)

Classical Free Electron Theory was proposed by Drude (1900) and extended by Lorentz. It treats the free electrons in a metal as a classical ideal gas obeying Maxwell–Boltzmann statistics, moving randomly and colliding with positive ions.

8.1 Assumptions

- Electrons in a metal behave as a classical ideal gas (free electron gas)
- Electron–electron and electron–ion interactions are neglected (free particle model)
- Electrons move randomly with average thermal speed related to temperature by: $\frac{1}{2}mv^2 = (3/2)k_BT$
- On applying an electric field, electrons drift opposite to the field with a drift velocity v_d
- Mean free path λ and relaxation time τ characterize collisions with ions

8.2 Derivation of Electrical Conductivity

1. Apply electric field E to the metal; force on electron: $F = -eE$
2. Equation of motion: $m\ dv/dt = -eE$
3. Between collisions (time τ): $v_d = -eE\tau/m$
4. Current density: $J = -nev_d = ne^2E\tau/m$
5. Since J = σE: $\sigma = ne^2\tau/m$

CLASSICAL CONDUCTIVITY FORMULAS

$$\sigma = ne^2\tau/m$$

σ = conductivity, n = electron density, e = charge, τ = relaxation time

$$\rho = 1/\sigma = m/(ne^2\tau)$$

Electrical resistivity

$$\mu = e\tau/m$$

Electron mobility; $J = ne\mu E$

$$v_d = -eE\tau/m = -\mu E$$

Drift velocity

8.3 Thermal Conductivity and Wiedemann–Franz Law

WIEDEMANN–FRANZ LAW

$$K/\sigma = (3/2)(k_B/e)^2\ T = L\cdot T$$

L = Lorenz number (classical) = $1.12 \times 10^{-8}\ W\Omega K^{-2}$

✓ Merits of Classical Theory

- Explains Ohm's law: $J \propto E$
- Derives expressions for electrical conductivity σ and resistivity ρ
- Explains Wiedemann–Franz law (ratio $K/\sigma T \approx$ constant)
- Explains metallic luster and general conducting behaviour of metals

✗ Demerits of Classical Theory

- **Specific heat failure:** Predicts electronic specific heat $C_v = (3/2)nk_B$, but observed value is ~100× smaller
- **Electrical conductivity at low T:** Cannot explain how conductivity increases at low temperatures
- **Hall coefficient:** Predicted value differs from experiment for many metals
- **Ferromagnetism:** Cannot explain ferromagnetism
- **Photoelectric effect:** Cannot explain photoelectric threshold independently
- **Mean free path:** Classical estimate gives ~1 Å, but experimental value is ~100–500 Å
- Does not distinguish between conductors, semiconductors, and insulators

9 Quantum Free Electron Theory (Sommerfeld Model)

Quantum Free Electron Theory was developed by Arnold Sommerfeld (1928). It treats conduction electrons quantum mechanically using Fermi–Dirac statistics and Pauli's exclusion principle, replacing the classical Maxwell–Boltzmann distribution.

9.1 Key Assumptions

- Electrons obey Schrödinger's wave equation (quantum mechanical treatment)
- Electrons are fermions: they obey Pauli's exclusion principle — no two electrons can occupy the same quantum state
- Energy distribution follows Fermi–Dirac statistics (not Maxwell–Boltzmann)
- The metal is treated as a potential box (electrons are free inside but confined within the metal)
- Ion cores create a uniform positive background (jellium model)

9.2 Energy of Free Electrons (Quantum Treatment)

Applying Schrödinger's equation to electrons in a 3D metallic box of side L:

QUANTUM ENERGY OF FREE ELECTRONS

$$E = (\hbar^2/2m)(k_x^2 + k_y^2 + k_z^2)$$

k = wave vector; $k_i = n_i\pi/L$

$$E = \hbar^2k^2/2m$$

$k^2 = k_x^2 + k_y^2 + k_z^2$

9.3 Comparison: Classical vs Quantum Theory

Property	Classical (Drude–Lorentz)	Quantum (Sommerfeld)
Statistics	Maxwell–Boltzmann	Fermi–Dirac
Energy distribution	Continuous	Discrete (quantized)

Specific heat	$C_v = (3/2)nk_B$	$C_v = \pi^2 n k_B^2 T/(2E_F)$
Lorenz number L	1.12×10^{-8} WΩK⁻²	2.44×10^{-8} WΩK⁻² ✓
Mean free path	~1 Å (wrong)	~100–500 Å (correct)
Temperature dependence of σ	Cannot explain T^{-1}	Explains correctly
Pauli exclusion	Not applied	Applied

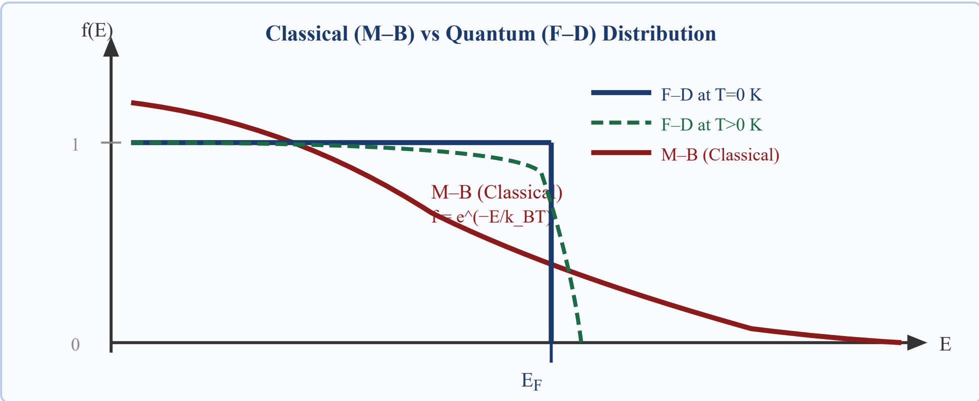

Fig. 9.1 — Classical (Maxwell–Boltzmann) vs Quantum (Fermi–Dirac) Energy Distribution

9.4 Merits of Quantum Theory

✓ Merits

- Correctly predicts the small electronic specific heat: $C_v \propto T$
- Explains correct Lorenz number: $L = 2.44 \times 10^{-8}$ WΩK⁻²
- Explains why only electrons near Fermi level participate in conduction
- Explains the large mean free path (~100–500 Å) observed experimentally
- Provides correct framework for band theory, semiconductors, and superconductors

10 Electrical Conductivity — Quantum Approach

Quantum Electrical Conductivity: In the quantum free electron model, only electrons with energy near the Fermi energy contribute to conduction. The expression for conductivity is similar in form to the classical result, but τ now represents the quantum relaxation time at E_F.

10.1 Derivation

1. In a metal under electric field E, only electrons near E_F are accelerated (others are blocked by Pauli exclusion)
2. The drift velocity acquired: $v_d = eE\tau_F/m$, where τ_F = relaxation time at Fermi level
3. Current density: $J = nev_d = ne^2\tau_F E/m$
4. Quantum conductivity: $\sigma = ne^2\tau_F/m$
5. Since $\tau_F = \lambda_F/v_F$ (mean free path at Fermi surface / Fermi velocity): $\sigma = ne^2\lambda_F/(mv_F)$

QUANTUM CONDUCTIVITY RELATIONS

$$\sigma = ne^2\tau_F/m$$

Quantum form; τ_F = relaxation time at Fermi level

$$\sigma = ne^2\lambda_F/(mv_F)$$

λ_F = mean free path, v_F = Fermi velocity

$$v_F = \hbar k_F/m = \sqrt{(2E_F/m)}$$

Fermi velocity (typically ~10^6 m/s)

$$\rho = m/(ne^2\tau_F) = \rho_{residual} + \rho_{lattice}(T)$$

Matthiessen's rule: total resistivity

10.2 Matthiessen's Rule

The total resistivity of a metal is the sum of contributions from temperature-dependent lattice vibrations (phonon scattering) and temperature-independent impurity/defect scattering:

MATTHIESSEN'S RULE

$$\rho = \rho_{thermal}(T) + \rho_{residual}$$

$$\rho_{thermal} \propto T \text{ (at high T)}$$

$$\rho_{thermal} \propto T^5 \text{ (at low T, Bloch-Grüneisen)}$$

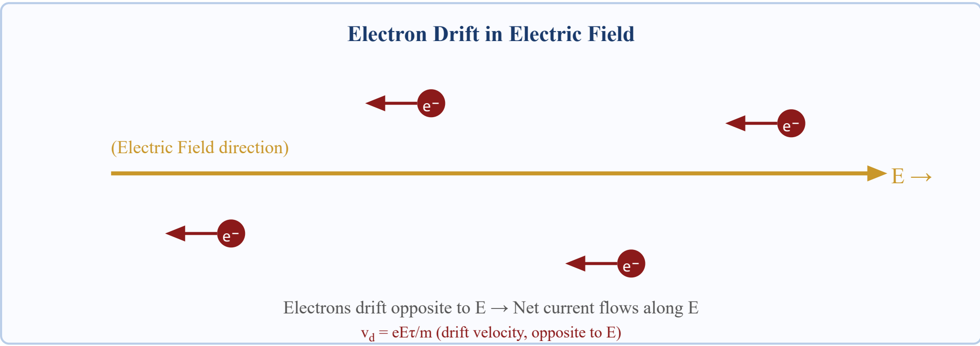

Fig. 10.1 — Electron Drift Motion Under Applied Electric Field

11 Fermi–Dirac Distribution

Fermi–Dirac Distribution: The statistical distribution that gives the probability of occupation of a quantum energy state E by an electron (a fermion) at thermal equilibrium temperature T.

FERMI–DIRAC DISTRIBUTION FUNCTION

$$f(E) = 1 / [e^{(E-E_F)/k_BT} + 1]$$

f(E) = probability of occupation of state with energy E

E_F = Fermi energy, k_B = Boltzmann constant, T = temperature

11.1 Behaviour at Different Temperatures

Case 1: At T = 0 K (Absolute Zero)

- For $E < E_F$: $f(E) = 1$ → All states below E_F are fully occupied
- For $E > E_F$: $f(E) = 0$ → All states above E_F are empty
- F–D function is a perfect step function at T = 0 K

Case 2: At T > 0 K (Finite Temperature)

- The step is "smeared" over an energy range ~k_BT around E_F
- Some electrons just below E_F are excited above E_F
- At $E = E_F$: $f(E_F) = 1/2$ always (for any T > 0)

Case 3: High Temperature ($E \gg E_F$)

- F–D distribution approaches Maxwell–Boltzmann: $f(E) \approx e^{-(E-E_F)/k_BT}$

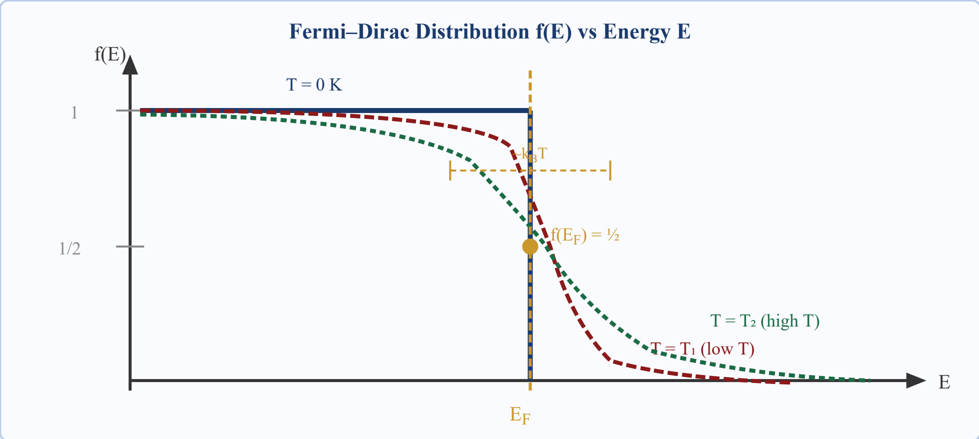

Fig. 11.1 — Fermi–Dirac Distribution: f(E) vs E at T=0 K and finite temperatures

11.2 Important Conclusions

- **At T = 0 K:** All states with $E < E_F$ are filled; all with $E > E_F$ are empty
- **At any T:** $f(E_F) = ½$ (Fermi energy is defined by this condition)
- **Smearing width:** Only electrons within $\sim k_BT$ of E_F are thermally active
- **Connects to classical:** When $(E - E_F) \gg k_BT$, $f(E) \longrightarrow$ Maxwell–Boltzmann

12 Density of States & Fermi Energy

12.1 Density of States g(E)

Density of States g(E): The number of quantum states available per unit energy per unit volume at energy E. It tells us how densely packed the energy levels are at a given energy.

12.1.1 Derivation of g(E)

1. In k-space, each electron state occupies a volume $(\pi/L)^3$
2. Number of states in a k-sphere of radius k: $N = 2 \times (4\pi k^3/3) / (2\pi/L)^3 = Vk^3/(3\pi^2)$
3. Since $E = \hbar^2k^2/2m \rightarrow k = \sqrt{(2mE)}/\hbar$
4. Differentiate N w.r.t. E: $g(E) = dN/dE = V(4\pi/h^3)(2m)^{3/2} \sqrt{E}$
5. Per unit volume: $g(E) = (4\pi/h^3)(2m)^{3/2} \sqrt{E}$

DENSITY OF STATES

$$g(E) = (4\pi(2m)^{3/2}/h^3) \sqrt{E}$$

Number of states per unit volume per unit energy at energy E

$$g(E) = (3n/2) E^{-3/2} \sqrt{E} = (3n/2E_F^{3/2}) \sqrt{E}$$

Alternative form using Fermi energy

$$g(E) \propto \sqrt{E}$$

Parabolic dependence on energy

12.2 Fermi Energy E_F

Fermi Energy E_F: The highest energy occupied by an electron at absolute zero (T = 0

K). It is the energy at which the Fermi–Dirac distribution equals ½.

12.2.1 Derivation of Fermi Energy at T = 0 K

1. At T = 0 K, all states up to E_F are filled: $n = \int_0^{E_F} g(E)\, dE$
2. Substituting g(E): $n = (4\pi(2m)^{3/2}/h^3) \int_0^{E_F} \sqrt{E}\, dE = (4\pi(2m)^{3/2}/h^3) \cdot (2/3)\, E_F^{3/2}$
3. Solving for E_F: $E_F^{3/2} = (3h^3 n)/(8\pi(2m)^{3/2})$
4. Therefore:

FERMI ENERGY FORMULAS

$$E_F = (\hbar^2/2m)(3\pi^2 n)^{2/3}$$

Primary formula; n = number density of electrons

$$E_F = (h^2/8m)(3n/\pi)^{2/3}$$

Equivalent form using h instead of ħ

$$E_F(T) \approx E_{F0}[1 - \pi^2 (k_B T)^2/(12 E_{F0}^2)]$$

Temperature correction (E_F slightly decreases with T)

$$k_F = (3\pi^2 n)^{1/3}$$

Fermi wave vector

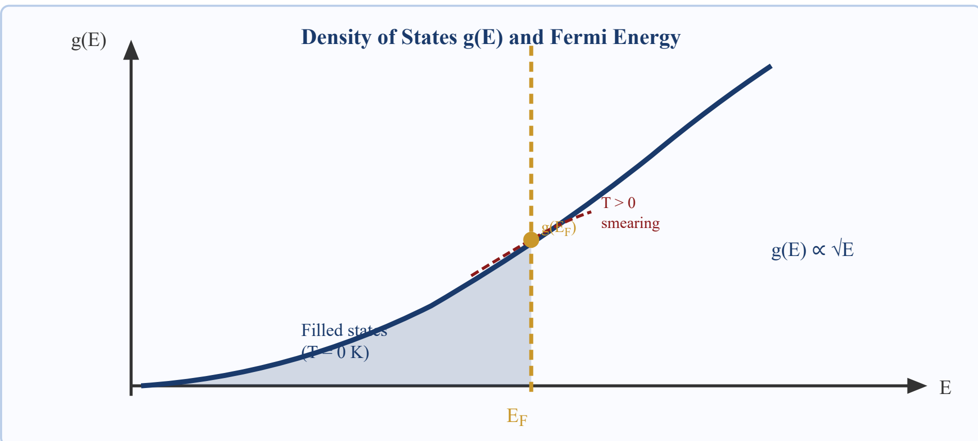

Fig. 12.1 — Density of States g(E) vs Energy E, with Fermi Energy E_F at T=0

12.3 Typical Fermi Energies

Metal	E_F (eV)	Fermi Temperature T_F (K)	Fermi Velocity v_F ($\times10^6$ m/s)
Copper (Cu)	7.04	81,600	1.57
Silver (Ag)	5.49	63,700	1.39
Gold (Au)	5.53	64,200	1.40
Aluminium (Al)	11.63	135,000	2.03
Iron (Fe)	11.16	129,000	1.98

12.4 Significance of Fermi Energy

- Separates filled and empty states at T = 0 K
- Determines which electrons participate in conductivity (those near E_F)
- Used to compute electronic specific heat, thermionic emission, and photoelectric emission
- Since $T_F \approx$ 10,000–100,000 K, room temperature (~300 K) is effectively "0 K" for electrons

13 Questions & Answers

Q1. What is de Broglie's hypothesis? State the expression for de Broglie wavelength. 2 Marks

de Broglie (1924) proposed that every moving particle, whether microscopic or macroscopic, is associated with a matter wave. Just as radiation has both wave and particle properties (photons), particles also exhibit wave-like behaviour under appropriate conditions.

The wavelength of this matter wave, called the **de Broglie wavelength**, is given by:

$$\lambda = h/p = h/mv$$

where h is Planck's constant (6.626×10^{-34} J·s), p = mv is the momentum of the particle. For a particle accelerated through potential V (electrons): **$\lambda = 12.27/\sqrt{V}$ Å**. This was verified by the Davisson–Germer experiment (1927) through electron diffraction from a nickel crystal.

Q2. Define Fermi energy. What is the value of $f(E_F)$ at any temperature? 2 Marks

Fermi Energy (E_F) is defined as the energy of the highest occupied quantum state by an electron at absolute zero temperature (T = 0 K). It represents the maximum energy that electrons can have at T = 0 K.

Mathematically: $\mathbf{E_F = (\hbar^2/2m)(3\pi^2 n)^{2/3}}$, where n is the electron number density.

At any finite temperature T, the Fermi–Dirac distribution gives: $\mathbf{f(E_F) = 1/[e^0 + 1] = 1/2}$. This means the Fermi energy is the energy at which the probability of occupancy is exactly ½ (50%) for all temperatures T > 0 K. For copper, $E_F \approx 7.04$ eV.

Q3. State Heisenberg's Uncertainty Principle. Write all three forms. 2 Marks

Heisenberg's Uncertainty Principle (1927) states that it is fundamentally impossible to simultaneously and precisely determine both the position and momentum of a quantum particle. The more precisely one quantity is determined, the less precisely the conjugate quantity can be known.

The three forms are:

$$\Delta x \cdot \Delta p_x \geq \hbar/2 = h/(4\pi) \qquad \text{(Position–Momentum)}$$

$$\Delta E \cdot \Delta t \geq \hbar/2 \quad \text{(Energy-Time)}$$

$$\Delta L \cdot \Delta\theta \geq \hbar/2 \quad \text{(Angular Momentum-Angle)}$$

This is NOT due to measurement error but is an intrinsic property of quantum nature arising from the wave character of matter.

Q4. Explain Heisenberg's Uncertainty Principle with a suitable example. Prove it using single-slit diffraction. 5 Marks

Statement: The product of the uncertainties in the position (Δx) and the momentum (Δp) of a quantum particle is always greater than or equal to $\hbar/2 = h/(4\pi)$:

$$\Delta x \cdot \Delta p \geq \hbar/2 = h/(4\pi)$$

Physical Basis: This arises from the wave nature of matter. A wave that is well-localized in space must contain many frequency components (broad momentum distribution). A wave with a well-defined frequency (definite momentum) must extend over all space (uncertain position). This is a mathematical result from Fourier analysis, not from experimental imprecision.

Proof using Single-Slit Diffraction:

Consider an electron beam passing through a slit of width d. The position uncertainty is $\Delta y = d$ (we know the electron must pass through this region).

Due to diffraction, the electron acquires transverse momentum. The first diffraction minimum occurs at angle θ where: $\sin\theta = \lambda/d$. The transverse momentum spread: $\Delta p_y = p \cdot \sin\theta = (h/\lambda) \cdot (\lambda/d) = h/d$.

Therefore: $\Delta y \cdot \Delta p_y = d \cdot (h/d) = h \approx \hbar$ (order of magnitude), confirming the uncertainty principle.

Example — Electron in a nucleus:

Nuclear radius $\approx 10^{-14}$ m, so $\Delta x \approx 10^{-14}$ m. From uncertainty principle: $\Delta p \geq \hbar/\Delta x = (1.05 \times 10^{-34})/(10^{-14}) \approx 10^{-20}$ kg·m/s. The kinetic energy corresponding to this momentum is ≈ 20 MeV. But observed β-particle energies are only ~1–2 MeV, so **electrons cannot reside inside the nucleus** — confirming the principle.

Applications: Zero-point energy of particles in boxes, spectral line broadening ($\Delta E \cdot \Delta t \geq \hbar$), stability of atomic orbits (electron cannot spiral into nucleus), and the principle of operation of the scanning tunnelling microscope (STM).

Q5. Derive the time-independent Schrödinger wave equation from first principles. 5 Marks

Starting Point: Consider a particle of mass m moving in region where potential energy is $V(x)$. The total energy is $E = KE + PE = p^2/2m + V$.

Step 1 — Wave function assumption: For a particle with definite energy E and momentum p, the wave function is: $\psi(x,t) = A\, e^{i(kx - \omega t)}$, where $k = p/\hbar$, $\omega = E/\hbar$.

Step 2 — Time differentiation: $\partial\psi/\partial t = -i\omega\cdot\psi = -i(E/\hbar)\psi$. Hence: $i\hbar\cdot\partial\psi/\partial t = E\cdot\psi$. This gives the *energy operator*: $\hat{E} = i\hbar\, \partial/\partial t$.

Step 3 — Spatial differentiation: $\partial^2\psi/\partial x^2 = (ik)^2\psi = -k^2\psi = -(p^2/\hbar^2)\psi$. Hence: $-\hbar^2\partial^2\psi/\partial x^2 = p^2\psi$, giving the *momentum operator* squared: $\hat{p}^2 = -\hbar^2\, \partial^2/\partial x^2$.

Step 4 — Energy equation: $E\psi = (p^2/2m + V)\psi$. Substituting operators: $i\hbar\partial\psi/\partial t = -(\hbar^2/2m)\partial^2\psi/\partial x^2 + V\psi$. This is the **Time-Dependent Schrödinger Equation (TDSE)**.

Step 5 — Separation of variables (for time-independent V): Let $\psi(x,t) = \varphi(x)\cdot e^{-iEt/\hbar}$. Substituting: $E\cdot\varphi\cdot e^{-iEt/\hbar} = [-\hbar^2/2m \cdot d^2\varphi/dx^2 + V\varphi]\cdot e^{-iEt/\hbar}$.

Step 6 — Cancelling $e^{-iEt/\hbar}$:

$$-(\hbar^2/2m)\ d^2\psi/dx^2 + V(x)\psi = E\psi$$

This is the **Time-Independent Schrödinger Equation (TISE)** in one dimension. In 3D: $-(\hbar^2/2m)\nabla^2\psi + V\psi = E\psi$, or equivalently: $\hat{H}\psi = E\psi$ where $\hat{H}$ is the Hamiltonian operator.

The solutions ψ are called **eigenfunctions** (wave functions) and E are the corresponding **eigenvalues** (allowed energy levels) of the quantum system.

Q6. Solve the Schrödinger equation for a particle in a one-dimensional infinite potential well. Obtain wave functions and energy levels. 5 Marks

Problem Setup: A particle of mass m is confined in a 1D box of length L. The potential: $V = 0$ for $0 \le x \le L$, and $V = \infty$ outside. Since $V = \infty$ outside, $\psi = 0$ outside.

Schrödinger Equation inside well (V = 0): $d^2\psi/dx^2 + (2mE/\hbar^2)\psi = 0$. Let $k^2 = 2mE/\hbar^2$, so $d^2\psi/dx^2 + k^2\psi = 0$.

General Solution: $\psi(x) = A \sin(kx) + B \cos(kx)$.

Boundary Conditions:

(i) At $x = 0$: $\psi(0) = 0 \rightarrow B \cos(0) = B = 0 \rightarrow B = 0$. So $\psi = A \sin(kx)$.

(ii) At $x = L$: $\psi(L) = 0 \rightarrow A \sin(kL) = 0 \rightarrow kL = n\pi$ ($n = 1, 2, 3,\ldots$). So $k_n = n\pi/L$.

Quantized Energy Levels: From $k^2 = 2mE/\hbar^2$: $E_n = \hbar^2k^2/2m = \hbar^2n^2\pi^2/(2mL^2) =$ $\mathbf{n^2h^2/(8mL^2)}$.

Normalization — Finding A: $\int_0^L|\psi|^2\,dx = 1 \rightarrow A^2\int_0^L \sin^2(n\pi x/L)\,dx = A^2 \cdot L/2 = 1 \rightarrow A = \sqrt{(2/L)}$.

Normalized Wave Function:

$$\psi_n(x) = \sqrt{(2/L)} \cdot \sin(n\pi x/L),\ n = 1, 2, 3, \ldots$$

Key Results: Energy is quantized ($E_n = n^2E_1$); zero-point energy $E_1 = h^2/(8mL^2) \neq 0$; wave function ψ_n has $(n-1)$ nodes; particle cannot be at rest inside the well due to zero-point energy.

Q7. Compare Classical and Quantum Free Electron Theories. Explain why the quantum theory is superior. 5 Marks

Classical Free Electron Theory (Drude–Lorentz, 1900–1909) treated conduction electrons as a classical ideal gas using Maxwell–Boltzmann statistics. It successfully derived Ohm's law and explained basic conductivity and the Wiedemann–Franz law qualitatively. However, it predicted an electronic specific heat of $C_v = (3/2)nk_B$, which is about 100 times larger than the experimentally observed value. It also predicted an incorrect Lorenz number (1.12×10^{-8} $W\Omega K^{-2}$), an unrealistically short mean free path (~1 Å vs observed ~100–500 Å), and failed to explain ferromagnetism, superconductivity, and the distinction between conductors, semiconductors, and insulators.

Quantum Free Electron Theory (Sommerfeld, 1928) applied quantum mechanics and Fermi–Dirac statistics to the electron gas. Key improvements:

(i) **Specific heat:** Only electrons within ~k_BT of E_F are thermally active ($\ll$1% of all electrons). So $C_v = \pi^2nk_B^2T/(2E_F) \propto T$, matching experiments.

(ii) **Lorenz number:** Quantum theory gives $L = 2.44 \times 10^{-8}$ $W\Omega K^{-2}$, in excellent agreement with experiment.

(iii) **Mean free path:** Quantum diffraction effects allow much longer mean free paths (~100–500 Å).

(iv) Correctly explains thermionic and photoelectric emission using the Fermi energy concept.

(v) Forms the foundation for band theory, semiconductors, and transistors.

Conclusion: Quantum theory replaces the classical Maxwell–Boltzmann distribution (which allows all electrons to contribute to transport) with the Fermi–Dirac distribution (where only electrons near the Fermi surface contribute). This single change resolves most failures of the classical theory and provides a far more accurate description of electron behaviour in metals.

Q8. Explain Fermi–Dirac distribution. Discuss its behaviour at T = 0 K and T > 0 K. Define density of states.

5 Marks

Fermi–Dirac Distribution: The probability that a quantum state of energy E is occupied by an electron at temperature T is given by:

$$f(E) = 1 / [e^{(E-E_F)/k_BT} + 1]$$

Behaviour at T = 0 K:

For $E < E_F$: The exponent → −∞, so f(E) = 1/(0 + 1) = 1. All states are fully occupied.

For $E > E_F$: The exponent → +∞, so f(E) = 1/(∞ + 1) = 0. All states are completely empty.

At T = 0 K, f(E) is a perfect step function: electrons fill all states up to E_F and none above it.

Behaviour at T > 0 K (finite temperature):

The step function is "smeared" over an energy range of approximately k_BT around E_F. Some electrons just below E_F are thermally excited to states just above E_F. At $E = E_F$: $f(E_F) = 1/[e^0+1] = 1/2$ (50% probability of occupation, for any T).

For $E - E_F \gg k_BT$, the Fermi–Dirac distribution reduces to the Maxwell–Boltzmann distribution: $f(E) \approx e^{-(E-E_F)/k_BT}$.

Density of States g(E): The number of quantum states available per unit energy per unit volume at energy E. For free electrons in 3D:

$$g(E) = (4\pi(2m)^{3/2}/h^3) \cdot \sqrt{E}$$

g(E) increases as $\sqrt{E}$ (parabolic). The actual number of electrons occupying states in [E, E+dE] is: n(E) = g(E) · f(E) · dE.

Significance: The product g(E)·f(E) gives the energy distribution of electrons. At room temperature, this distribution is almost identical to the T=0 K distribution for metals, since $k_BT \approx 0.025$ eV $\ll E_F \approx 5$–10 eV for typical metals.

14 Quick Revision Sheet — Unit IV

⚛ QUANTUM MECHANICS FORMULAS

$E = h\nu = \hbar\omega$ [Planck]

$\lambda = h/p = h/mv$ [de Broglie]

$\lambda = 12.27/\sqrt{V}$ Å [Electrons]

$\Delta x \cdot \Delta p \geq \hbar/2$ [Heisenberg]

$\Delta E \cdot \Delta t \geq \hbar/2$ [Energy-Time]

$\int|\psi|^2\, dV = 1$ [Normalisation]

📐 SCHRÖDINGER EQUATIONS

$i\hbar\partial\psi/\partial t = \hat{H}\psi$ [TDSE]

$\hat{H}\psi = E\psi$ [TISE]

$-(\hbar^2/2m)d^2\psi/dx^2+V\psi=E\psi$

$\psi_n=\sqrt{(2/L)}\ \sin(n\pi x/L)$

$E_n = n^2h^2/(8mL^2)$ [Box]

$E_1 = h^2/(8mL^2)$ [ZPE]

🔋 FREE ELECTRON THEORY

$\sigma = ne^2\tau/m$ [Classical]

$v_d = eE\tau/m$ [Drift vel.]

$\mu = e\tau/m$ [Mobility]

$J = \sigma E = ne\mu E$ [Ohm's law]

$\rho = \rho_{res}+\rho_{thermal}$ [Matthiessen]

$K/\sigma T = L$ [Wiedemann-F]

📊 FERMI–DIRAC & G(E)

$f(E)=1/[e^{(E-E_F)/k_BT}+1]$

$f(E<E_F)=1$ @ $T=0K$

$f(E>E_F)=0$ @ $T=0K$

$f(E_F)=½$ @ any T

$E_F=\hbar^2(3\pi^2n)^{2/3}/(2m)$

$g(E)\propto\sqrt{E}$ [DOS parabola]

Concept Summary

Concept	Key Idea	Formula/Result
de Broglie	Particles have wave nature	$\lambda = h/mv$
Heisenberg UP	Position & momentum cannot both be exact	$\Delta x\cdot\Delta p \geq h/4\pi$
Wave function ψ	$|\psi|^2$ = probability density	$\int|\psi|^2dV = 1$
TDSE	Time evolution of quantum state	$i\hbar\, \partial\psi/\partial t = \hat{H}\psi$
TISE	Stationary states, energy levels	$\hat{H}\psi = E\psi$
Particle in box	Confinement → discrete energies	$E_n = n^2h^2/8mL^2$

Classical FET	M-B statistics; explains Ohm's law	$\sigma = ne^2\tau/m$
Quantum FET	F-D statistics; correct C_v, Lorenz	$C_v \propto T$
Fermi–Dirac	Probability of electron occupancy	f(E) = 1/[e^{(E−E_F)/kT}+1]
Density of States	States per unit energy per unit volume	$g(E) \propto \sqrt{E}$
Fermi Energy	Highest filled energy at T = 0 K	E_F = ħ²(3π²n)^{2/3}/2m

Important Derivations to Remember

- **de Broglie wavelength:** From $E = h\nu$ and $E = pc \rightarrow \lambda = h/p$ (by analogy with photon)
- **Heisenberg UP (single slit):** $\Delta y = d$, $\Delta p = h/d \rightarrow \Delta y \cdot \Delta p = h$ (order of ħ)
- **TISE derivation:** Start from de Broglie wave → differentiate → apply energy conservation
- **Particle in box:** Apply TISE with V=0, boundary conditions $\psi(0)=\psi(L)=0 \rightarrow E_n = n^2h^2/8mL^2$
- **Classical conductivity:** Newton's law + drift → $\sigma = ne^2\tau/m$ (Drude model)
- **Fermi energy:** Integrate g(E) from 0 to E_F = n → E_F = (ħ²/2m)(3π²n)^{2/3}

Key Graph Summaries

Graph	Shape	Key Feature
$\psi_n(x)$ in box	Sinusoidal (n half-waves)	n−1 nodes; zero at boundaries
$\|\psi_n\|^2$ in box	sin² pattern (always ≥ 0)	n peaks; $\|\psi_1\|^2$ → uniform at large n
Energy levels in box	Discrete horizontal lines, spacing ∝ n²	$E_n = n^2E_1$; levels spread apart
F–D at T=0	Perfect step at E_F	f=1 below E_F; f=0 above
F–D at T>0	S-shaped smeared step	$f(E_F)$ = ½ always
g(E) vs E	Parabola (√E)	More states at higher energy
σ vs T	Decreasing (σ ∝ 1/T for metals)	Phonon scattering increases with T

Physical Constants

Constant	Symbol	Value
Planck's constant	h	6.626×10^{-34} J·s
Reduced Planck's constant	$\hbar = h/2\pi$	1.055×10^{-34} J·s
Electron mass	m_e	9.109×10^{-31} kg
Electron charge	e	1.602×10^{-19} C
Boltzmann constant	k_B	1.381×10^{-23} J/K
Speed of light	c	3×10^{8} m/s

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